Algorithmic Impact Assessment Results

Version: 0.10.0

Project Details

1. Name of Respondent
Chantal Demers

2. Job Title
Manager

3. Department
Canada Border Services Agency

4. Branch
Travellers

5. Project Title
Traveller Compliance Indicator

6. Departmental Program (from Department Results Framework)
Traveller Facilitation and Compliance

7. Project Phase
Design [Points: 0]

8. Please provide a project description:

The Canada Border Services Agency (CBSA) provides integrated border services to protect national security and public safety while facilitating the lawful flow of people and goods across our border.

Each year the CBSA welcomes more than 80 million travellers. New technologies are being implemented to provide a more efficient border experience; including tools that incorporate artificial intelligence, as part of the CBSA's Traveller Modernization Initiative. The CBSA is committed to privacy. All new technologies must respect Canada's privacy policies to protect personal information and ensure it's used responsibly.

The Traveller Compliance Indicator (TCI) is a new tool that will provide more information to border services officers (BSOs) to help them identify travellers that meet customs-related obligations. The TCI uses historical data available in CBSA systems, in addition to live traveller data available at the time of entry into Canada at the port of entry. The tool is expected to save travellers time by reducing the number of non-resultant secondary inspections while allowing officers to focus their attention on travellers who present a higher risk of non-compliance with border legislation and associated regulations.

The CBSA has made investments over the last few years to develop, test (i.e., offline tests were run retrospectively using the completed passages) and refine the TCI algorithm and scoring models to identify travellers, who based on their previous travel activity, are more likely to be in compliance with the *Customs Act.*

Methodologically, the TCI process begins with data pre-processing, where data from two separate CBSA systems are linked: Passage History (i.e., information from Primary Inspection Lanes (PIL) and the Integrated Customs Enforcement System (ICES) (i.e., enforcement records). Next, feature engineering is applied to enhance the dataset with additional features that help identify compliant passages. Feature engineering refers to the process of preparing and transforming raw data into meaningful features (variables or attributes) that can improve the performance of machine learning models. In this context, the new features capture travel patterns, such as frequency, locations, and times of travel. Traveller names or IDs are necessary to track historical passages; however, only data from Passage History and ICES is used for this purpose.

After feature engineering, the data is de-identified to protect privacy. Personally identifiable information (names, dates of birth, IDs, document numbers, and exact date/time of passage, etc.) is removed. Instead of precise ages, travellers are grouped into four-year age categories. Sex and nationality are retained for operational decision-making, and with all other identifying details removed, the dataset is considered anonymized.

The de-identified dataset is then used to train a machine learning model. The algorithm, structured as a decision tree, learns general patterns to predict compliance for new passages. For example, it may learn that travellers over 60 who travel later in the year are typically compliant. The model is tested to ensure it generalizes well and does not rely on simplistic or biased patterns. Some of the steps to mitigate bias within the data set include using random referrals within the training data sets to mitigate the reliance solely on secondary referrals by BSO, or those generated by targets and intelligence lookouts. The approach uses explainable artificial intelligence, so the decision logic can be easily understood and analyzed.

When deployed, the model operates in real time during a traveller's passage at the border. The system retrieves the traveller's history, calculates features, and predicts compliance. If a high likelihood of compliance is predicted, an indicator is displayed to the officer.

The data features used to train the TCI models will be reviewed both in response to the active and continual monitoring of the model's performance, and as the CBSA data scientists create new features with the aim of boosting predictive power and mitigating against potential biases and performance degradation over time.

The TCI outputs will be communicated through the internal operating system used by BSOs at border crossings, in order to support their decision-making process and improve referral to secondary examination practices. The TCI system produces two types of outputs on the primary processing screen for a traveller that is available for the BSO to consider: a high likelihood of compliance, indicated by an indicator, or no icon, which reflects unknown risk. The TCI system will work concurrently with other CBSA traveller systems during the primary inspection process with the same level of quick response times so as to not impede the CBSA's objective of efficient traveller flow while also providing BSOs with improved access to the historical passage data at the PIL. While the TCI measures travellers that are highly likely to be in compliant, there are other CBSA systems that are presently in place to flag high risk travellers, namely enforcement and intelligence systems, and in these instances the traveller will be referred for secondary examination and any TCI output will be overridden. The TCI will be implemented within the new Officer Experience (OE) solution in all modes for in-person processing channels using a phased approached according to the OE project schedule. Initial deployment will be at CBSA ferry and land border crossings starting in 2026.

The CBSA's Traveller Modernization Initiative aims to develop new tools and technologies for a faster, better and secure experience at the border.

The CBSA is in the process of a major transformation, driven by the need to modernize how front-line services are delivered and operations are managed. The intent is to modernize the CBSA's existing risk-based approach to border management by investing in digital solutions to strengthen its capacity for evidence-based decision making and improve performance management. This is reflected in the CBSA's Data Analytics Strategy, resulting in data that is of a high quality, accessible, and well managed and that supports effective business intelligence and risk management.

The OE Initiative involves the consolidation and rationalization of multiple operational systems, the migration of data to the hybrid cloud environment, use of analytics to support officer decision-making, and the development of a single, user-friendly interface to support officer decisions and their operational needs. To ensure border management keeps pace with a changing world, the CBSA is modernizing services and processes to meet the increasing numbers of travellers. The CBSA understands that new technologies may raise concerns about privacy, bias and protection of personal information. To address this, the CBSA is committed to ensuring it develops its technologies in accordance with applicable legal frameworks, including ensuring compliance with the *Privacy Act* and the *Charter*.

At the border, the processing of a traveller sometimes includes secondary examinations, a routine part of traveller processing, including the examination of travellers and goods, at Canadian POEs to ensure compliance with Canada's customs, immigration, food, plant and animal laws and regulations. The CBSA has traditionally supported front-line operations with enforcement and intelligence processes, in addition to equipping officers with detection technology and enforcement tools in order to strengthen capacity to intercept traveller non-compliance at POEs. The TCI provides the ability to assess whether a traveller is highly likely to be compliant by assessing previous passage history and secondary referral results. As such, the CBSA is not introducing the TCI as an enforcement feature but rather as a tool to support traveller facilitation during the border clearance process.

TCI Benefits
The project is expected to:

- Allow officers to focus limited resources on travellers who the TCI determines' risk levels are unknown or high-risk.
- Improve the accuracy of referrals and further process travellers more efficiently, including leveraging data already available in CBSA systems.
- Speed up processing of travellers with passage characteristics that are consistent with compliant trends.

The TCI will move the CBSA to a risk-based compliance model to allow for fewer, and faster interactions with lower-risk travellers, while increasing examination capacity for high and unknown-risk travellers. The TCI data elements that were included within the algorithm were chosen by using 5 years of passage history and enforcement records from ICES, which were then analyzed to identify patterns and trends of travellers entering Canada. Examples of historical data include number of passages a traveller has over a specific timeframe; most common time of day for a traveller's passage; a traveller's most frequently crossed of entry.

About The System

9. Please check which of the following capabilities apply to your system.
Other (please specify)

10. Please describe.
Triage- Based on identified patterns, TCI focuses on predicting compliance of travellers and highlighting those who are highly likely to be compliant (i.e., low-risk travellers). In all cases, no action is triggered automatically, and officers always retain discretion to make the final decision based on the totality of the information available to them to release or refer the traveller for secondary processing.

In cases where systems (other than TCI) that identify potentially high-risk travellers are triggered, the traveller will be flagged for referral to secondary for further examination. In these instances, the officer will follow the required protocol prior to proceeding with a referral for additional processing in secondary, whether for customs and/or immigration secondary processing subject to the officer's judgment and discretion.

Section 1: Impact Level : 2

Current Score: 45

Raw Impact Score: 45

Mitigation Score: 32

Section 2: Requirements Specific to Impact Level 2

Peer review

Consult at least one of the following experts and publish the complete review or a plain language summary of the findings on a Government of Canada website:

- qualified expert from a federal, provincial, territorial or municipal government institution
- qualified members of faculty of a post-secondary institution
- qualified researchers from a relevant non-governmental organization
- contracted third-party vendor with a relevant specialization
- a data and automation advisory board specified by Treasury Board of Canada Secretariat.

OR

Publish specifications of the automated decision system in a peer-reviewed journal. Where access to the published review is restricted, ensure that a plain language summary of the findings is openly available.

Gender-based Analysis Plus

Ensure that the Gender-based Analysis Plus addresses the following issues:

- impacts of the automation project (including the system, data and decision) on gender and/or other identity factors;
- planned or existing measures to address risks identified through the Gender-based Analysis Plus.

Notice

Plain language notice posted through all service delivery channels in use (Internet, in person, mail or telephone).

Human-in-the-loop for decisions

Decisions may be rendered without direct human involvement.

Explanation

In addition to any applicable legal requirement, ensure that a meaningful explanation is provided to the client with any decision that results in the denial of a benefit or service, or involves a regulatory action. The explanation must inform the client in plain language of:

- the role of the system in the decision-making process;

- the training and client data, their source, and method of collection, as applicable;
- the criteria used to evaluate client data, and the operations applied to process it;
- the output produced by the system and any relevant information needed to interpret it in the context of the administrative decision; and
- a justification of the administrative decision, including the principal factors that led to it.

Explanations must also inform clients of relevant recourse options, where appropriate.

A general description of these elements must also be made available through the Algorithmic Impact Assessment and discoverable via a departmental website.

Training

Documentation on the design and functionality of the system.

IT and business continuity management

None

Approval for the system to operate

None

Other requirements

The Directive on Automated Decision-Making also includes other requirements that must be met for all impact levels.

[Link to the Directive on Automated Decision-Making]

Contact your institution's ATIP office to discuss the requirement for a Privacy Impact Assessment as per the Directive on Privacy Impact Assessment.

Section 3: Questions and Answers

Section 3.1: Impact Questions and Answers

Reasons for Automation

1. What is motivating your team to introduce automation into this decision-making process? (Check all that apply)
Improve overall quality of decisions
Lower transaction costs of an existing program
Use innovative approaches
Other (please specify)

2. Please describe

The TCI system uses data that is derived from data already in CBSA databases and is processed in real-time so as to allow the officer to focus on their interview of the traveller during the primary inspection process instead of having to undertake manual data searches that are inefficient and may lead to increased border processing times and may also lead to distraction and errors that negatively impact border security. As a result of faster, and more accurate traveller processing, it is expected that increases in traveller volume would not necessitate a relative increase in BSOs. Further, it would allow for BSOs to focus their time on processing travellers who are believed to present a high or unknown risk of non-compliance from a *Customs Act* and associated regulations perspective.

3. What client needs will the system address and how will this system meet them? If possible, describe how client needs have been identified.

All travellers, including Canadian citizens and permanent residents, expect to be treated fairly and professionally during the border clearance process. They also expect the CBSA to facilitate the efficient movement of persons and goods, consistent with the CBSA's mandate set out in the *CBSA Act*. In practical terms, this translates into a need for timely, predictable, and efficient processing, particularly for recurring travellers who have previously entered Canada. These needs have been identified through operational experience, observations of traveller volumes and wait times, and the CBSA's broader efforts to improve client service and facilitation at the border.

Although the TCI is not a public facing tool and is only available to BSOs, it is designed to support these same client needs. By using available historical data to identify travellers whose passage characteristics are consistent with compliant trends, the TCI helps BSOs focus their attention on travellers who may present a higher or unknown risk of noncompliance. This supports more efficient processing of compliant travellers and helps avoid unnecessary referrals to secondary examination, thereby enhancing fairness, consistency, and overall facilitation in the border clearance process.

By providing BSOs with the TCI at the initial contact stage, it will allow BSOs to spend less time with travellers believed to be highly likely to be compliant and more time on travellers that may pose a high or unknown risk. Spending less time on travellers believed to be highly likely compliant will result in reduced processing times for lower-risk travellers and support the core of CBSA's mandate which is to provide integrated border services that support national security and public safety priorities and facilitate the free flow of persons and goods.

4. Please describe any public benefits the system is expected to have.

The TCI will allow BSOs will be able to spend less time with compliant travellers and focus their attention on travellers who present a higher risk of non-compliance with border legislation and associated regulations.

5. How effective will the system likely be in meeting client needs?
Very effective [Points: +0]

6. Please describe any improvements, benefits, or advantages you expect from using an automated system. This could include relevant program indicators and performance targets.

The TCI system will improve client service through potential time savings through the use of a compliance prediction tool that leverages data already available in CBSA systems. A traveller's time spent with BSOs will decrease, on average, due to quicker access of existing records resulting in an improved capability of BSOs to make informed decisions at Primary thus reducing non-resultant secondary referrals by up to 20-30 percent. Critical Success Factors, Key Performance Indicator (KPI) and Success Criteria were used to determine the measurable outcomes (Efficiency, Usability and Effectiveness) of the TCI. Outcomes from previous offline tests were analyzed and it was concluded that they achieved the predetermined success criteria for the project. The analysis demonstrated the system's potential to help reduce the overall selective non-resultant referral rate by identifying compliant travellers with a high degree of accuracy.

In summary, some of the expected benefits are:
1) Ability to focus limited resources on unknown and higher risk travellers

2) Improve the accuracy of referrals
3) Speed up processing of lower risk travellers

7. Please describe how you will ensure that the system is confined to addressing the client needs identified above.

The CBSA will ensure that the TCI system remains confined to the client needs identified above through its defined program scope and supporting business requirements. These documents establish that the purpose of the TCI is to support efficient border processing for travellers entering Canada and to reduce unnecessary referrals for secondary processing by identifying travellers who are likely to be compliant. Any proposed change to the TCI system, including changes to its scope, data inputs, or outputs, is subject to existing CBSA program management and governance processes. These processes require that changes be assessed against the CBSA's mandate and ongoing client service objectives and be supported by the necessary analysis, including legal, privacy, and data management considerations informed by relevant policy instruments (e.g., the Directive on Automated Decision Making and the Directive on Service and Digital).

8. Please describe any trade-offs between client interests and program objectives that you have considered during the design of the project.

Several trade-offs between client interests and program objectives were considered in the design of TCI. First, by indicating characteristics that, the presence of which tend to predict a higher likelihood of compliance for up to approximately 30% of all travellers, the system, supports the program objective of improving facilitation and operational efficiency. The TCI decreases processing time for some travellers that are indicated to be highly likely compliant and provides BSOs with additional information to support their release and referral decisions. Recognizing lower risk travellers at primary can reduce non-resultant referrals to secondary examination, which saves time for both travellers and BSOs.

Second, using existing traveller data to create a new tool to be used during examinations may be perceived by some travellers as a trade-off between their interest in limiting new uses of their data and the program's objective of improving efficiency and targeting. While the TCI relies on data already available in CBSA systems, and the system is designed to preserve privacy and integrity, some travellers may still view this re-use of data as a new form of data processing. The CBSA will ensure that the collection, use, and disclosure of traveller information for the purposes of the TCI is consistent with its authorities under the *Customs Act* and complies with applicable obligations under the *Privacy Act*. The program's objective, however, is to use available data more effectively to support more efficient and consistent processing for all travellers.

Third, another trade-off would be related to resource allocation as it is anticipated that there will be an increase of travellers in the coming years.
Therefore, if our current processes remain unchanged, additional BSOs would be required to support frontline operations. The TCI feature will contribute to faster processing times and allow the CBSA to continue operating successfully with our current workforce complement. It is also to be noted that the system will not be replacing BSOs or their decision-making but instead allowing them to focus on travellers who may present an unknown or higher risk of non-compliance.

Finally, because TCI represents the CBSA's first use of artificial intelligence in a real-time at-border setting, a deliberate design choice was made to prioritize explainability over the predictive power of the trained models. In other words, the ability to explain the logic of the TCI models in granular detail was favoured over more complex modelling techniques that might have increased the proportion of passages resulting in a indicator. This decision aims to build trust in the project both internally and with the travelling public.

9. Have alternative non-automated processes been considered?
Yes [Points: +0]

10. If non-automated processes were considered, why was automation identified as the preferred option?

The current primary inspection process at land border crossings consists of a standard suite of automated system assessments for all travellers, If during the primary inspection process an officer decides to supplement the existing automated system results with additional searches on a traveller, there is currently only one option available to the officer - conducting manual searches of traveller information within CBSA systems. This is time consuming, inefficient and distracting.

Automation was identified as the preferred option for the new TCI feature as it provides a more organized and synthesized analysis of the data, more quickly than would be available in a non-automated process, again avoiding reliance on manual searches by officers that can also be subject to error due to inaccurate input of a travellers biographic information. It will allow BSOs to focus on higher risk travellers instead of travellers predicted to be highly likely compliant. Also, it will allow the CBSA to keep the primary processing border wait time service level at the current rate in the event that forecasted higher volume become reality.

Additionally, using automation will therefore allow the following benefits that would not be possible without the system:

1. Increasing overall efficiency of traveller processing by ensuring a quicker processing of travellers
2. Reducing transactional cost associated to traveller processing will result in cost avoidance
3. Increasing overall effectiveness of traveller processing

11. What would be the consequence of not deploying the system?
Service will be delivered in a less timely and efficient manner. [Points: +0]

Risk Profile

12. Is the project within an area of intense public scrutiny (e.g. because of privacy concerns) and/or frequent litigation?
Yes [Points: +3]

13. Are clients in this line of business particularly vulnerable?
Yes [Points: +3]

14. Are stakes of the decisions very high?
No [Points: +0]

15. Will this project have major impacts on staff, either in terms of their numbers or their roles?
No [Points: +0]

16. Will the use of the system create or exacerbate barriers for persons with disabilities?
No [Points: +0]

Project Authority

17. Will you require new policy authority for this project?
No [Points: +0]

About the Algorithm

18. The algorithm used will be a (trade) secret

Yes [Points: +3]

19. The algorithmic process will be difficult to interpret or to explain
No [Points: +0]

About the Decision

20. Please describe the decision(s) that will be automated.
The TCI outputs are not a decision. Rather they are indicators that may be used in the decision-making process by an officer to release a traveller or to continue the examination, such as with a referral for continued examination in secondary.

21. Does the decision pertain to any of the categories below (check all that apply):
Access and mobility (security clearances, border crossings) [Points: +1]

Impact Assessment

22. Which of the following best describes the type of automation you are planning?
Partial automation (the system will contribute to administrative decision- making by supporting an officer through assessments, recommendations, intermediate decisions, or other outputs). [Points: +2]

23. Please describe the role of the system in the decision-making process.

TCI will provide a predicted compliance indicator for each traveller based on an algorithmic assessment. The TCI outputs leverage historical data in CBSA systems and provide additional information to BSOs to support final decision on whether to release a traveller for entry to Canada or refer the traveller for secondary examination.

Again, it is to be noted that the system is not making any decisions on its own but is an additional tool provided to BSO to aid in their decision-making. The final decision to release or refer a traveller to secondary will always be made by the BSO, as is the case today, and is based on a multitude of factors, including the BSOs interview of the traveller.

24. Will the system be making decisions or assessments that require judgement or discretion?
Yes [Points: +4]

25. Please describe the criteria used to evaluate client data and the operations applied to process it.

The initial phases of the TCI project included a lengthy qualitative assessment of all data being proposed as necessary to the training of the TCI models. This included an assessment of the quality and completion of the data, and assessments from the front-line stand of its relative importance in the operational setting (i.e., how the front-line staff view the utility of such data). During this initial period, the project team, data scientist and groups of BSOs were tightly integrated in order to collaboratively identify data features captured in the CBSA's source systems, and others that could be derived from the source data, that could be of use in identifying compliant behaviour.

In order to action these insights and create the dataset to be used for model training, a record linkage process was needed to link CBSA transaction records to the resulting enforcement records. This process was scrutinized in detail to ensure the accuracy of the linkages and will continue to be monitored as the CBSA modernizes its identity resolution capabilities. This analysis will ensure that accurate traveller histories can be established, and that any patterns and

trends derived from such data accurately reflect reality.
With the collected and linked data, the TCI models could be trained and analyzed for performance against various metrics that aim to establish the accuracy of their performance and their performance across sociodemographic groupings in order mitigate against bias. For example, these metrics include:

- The percentage of passages receiving the indicator.
- The percentage of passages that have been examined and found to be compliant who received the indicator.
- The percentage of passages for which a customs enforcement action was issued that received the indicator.

Dashboards and pipelines have been created as part of the project's IT infrastructure to be monitored in near real-time in order to assure that TCI is functioning as expected, thus ensuring that any implementation or data related problems will be rapidly detected and addressed.

26. Please describe the output produced by the system and any relevant information needed to interpret it in the context of the administrative decision.

The TCI algorithm can produce an output that BSOs can consider in the context of making their administrative decisions. The system can display an indicator when the model predicts that the current traveller matches a compliance pattern. This means that the traveller matches to one of the "lower-risk" patterns, accordingly, the TCI output would generate a indicator and if the score is higher than the threshold value, then it does not display a indicator. This means that a traveller's predicted compliance is "not known" or below the threshold value.

At the time, TCI functionality is deployed to view on screen as a part of the OE project, the TCI assessment will be triggered by the Primary passport scan, license plate scan or Advance Declaration submissions. This may help focus BSOs questions when making their decision to Release or refer the traveller to secondary examination. Training for BSOs is being discussed in working groups and training materials will be provided before deployment.

27. Will the system perform an assessment or other operation that would not otherwise be completed by a human?
No [Points: +0]

28. Is the system used by a different part of the organization than the ones who developed it?
Yes
[Points: +4]

29. Are the impacts resulting from the decision reversible?
Reversible [Points: +1]

30. How long will impacts from the decision last?
Impacts are most likely to be brief [Points: +1]

31. Please describe why the impacts resulting from the decision are as per selected option above.

The TCI will provide more information to BSOs to assist them in identifying travellers with passage characteristics that are consistent with compliant trends. Although testing has demonstrated that TCI output predictions are correct 99 percent of the time, there is a slight chance that the compliance prediction is inaccurate. In all cases, BSOs make decisions based on an examination, visual indicators and the information made available to them when matching the traveller's identification to their entry records. The BSO could refer a traveller to secondary or release a traveller regardless of the TCI result. If a traveller is referred, the typical time spent in secondary is dependent on multiple factors including the type of exam. While statistics are collected to determine the time that passes between the time the traveller is referred to secondary from primary and the time of completion of the secondary examination, it may not be reflective of the actual exam time as this is dependent on when the officer actually updates the secondary examination system to reflect the examination status as closed.

Releasing travellers that were non-compliant and should have been referred to secondary examination could have a negative impact on national security, economic interests and health and safety. However, it should be noted this will only be decreasing those risks as it will be providing an additional tool to BSOs to make more informed decisions. Furthermore, referral results will continue to be used to refine the models. TCI outputs are kept for the sole purposes of system analysis and are not used when calculating subsequent passages by the same traveller.

32. The impacts that the decision will have on the rights or freedoms of individuals will likely be:
Little to no impact [Points: +1]

33. Please describe why the impacts resulting from the decision are as per selected option above.

The TCI system should have little to no impact on the rights or freedoms of travellers as the TCI only provides compliance-related predictive information to BSOs derived from existing data already held in passage history and ICES. As previously reiterated, the TCI system does not make any decisions and represents additional information provided to BSOs to assist them with their decision-making process. The final decision and direction given to a traveller for entry to Canada or for referral to secondary examination will always be made by the BSO not the TCI system.

Despite the one percent of cases where the TCI output is “incorrect” when predicting whether a traveller is highly likely to be compliant, it is only one element among a multitude of elements that can be used to support a BSOs decision to refer or release a traveller. BSOs are highly skilled and trained to detect potential instances of non-compliance during the primary inspection process therefore will always use these skills and training to support their final decision.

34. The impacts that the decision will have on the equality, dignity, privacy, and autonomy of individuals will likely be:
Little to no impact [Points: +1]

35. Please describe why the impacts resulting from the decision are as per selected option above.

The TCI system should have little to no impact on equality, dignity or privacy of travellers as the TCI only provides compliance-level information to BSOs based on the data already included in other systems. However, since data is used in new ways, travellers may not understand how the information is treated and how it may impact processing when arriving in Canada. In this regard, the system will gather information derived from Passage History and ICES, modify it, which will in turn will provide a compliance score for each traveller based on an algorithmic assessment. This information will provide additional information to the BSO to make their decision and will be used to expedite the process at the borders. With this data, the BSOs will be able to make more informed decisions in a reduced period of time. If the system was not at the disposition of the BSOs, they would still be able to perform an analysis based on all the data they currently have access to.

As previously stated, the TCI system does not make any decisions and represents additional information provided to BSOs to assist them with their decision-making process. The final decision and direction given to a traveller for entry to Canada or for referral to secondary examination will always be made by the BSO not the TCI system.

Additional assessments (Privacy Impact Assessment & GBA Plus Assessment) have been completed to ensure all factors related to privacy, equality and dignity of individuals are taken into consideration and therefore this is why the impact should be minimal.

36. The impacts that the decision will have on the health and well-being of individuals will likely be:
Little to no impact [Points: +1]

37. Please describe why the impacts resulting from the decision are as per selected option above.

The TCI system should have little to no impact on health and well-being of travellers as the TCI only provides additional information to BSOs to consider, derived from data already available in other systems.

As previously stated, the TCI system does not make any decisions and represents additional information provided to BSOs to assist them with their decision-making process. The final decision and direction given to a traveller for entry to Canada or for referral to secondary examination will always be made by the BSO not the TCI system.

Additionally, the tool will be tested quarterly in the backend systems before it is available and used in real-time. This will also contribute to supporting the tool's accuracy and ensuring BSOs know how to use it to support travellers best interests in regards to health and well-being.

38. The impacts that the decision will have on the economic interests of individuals will likely be:
Little to no impact [Points: +1]

39. Please describe why the impacts resulting from the decision are as per selected option above.

The TCI system should have little to no impact on economic interests as the TCI only provides compliance-related information to BSOs based on the data derived from other systems. BSOs will still perform an analysis of all the information available and the final decision to release or refer will be at their discretion.

In regard to the economic interests of individuals, the system could be favourable. As the processing times are expected to be reduced by the introduction of the TCI, it will expedite the process for travellers while improving the likelihood of crossing the borders without any unnecessary delays. Better processing times would contribute to seamless processes reducing unnecessary delays that should have a positive impact on other travel arrangements. Additionally, increased efficiencies in border processing will better prepare the CBSA to manage higher travel volumes without necessitating a relative increase in BSOs. This is why the impact should be minimal and will support travellers best interests in regards to economic interests.

As previously stated, the TCI system does not make any decisions and represents additional information provided to BSOs to assist them with their decision-making process. The final decision and direction given to a traveller for entry to Canada or for referral to secondary examination will always be made by the BSO not the TCI system.

40. The impacts that the decision will have on the ongoing sustainability of an environmental ecosystem, will likely be:
Little to no impact [Points: +1]

41. Please describe why the impacts resulting from the decision are as per selected option above.

As the system is data driven, it does not have a significant impact on ongoing sustainability of an environmental ecosystem. The system has little to no impact on environmental ecosystem as it is operational in a digital environment, however, reducing border wait times could have a very small impact on fuel emissions and therefore the environment.

About the Data - A. Data Source

42. Will the Automated Decision System use personal information as input data?
No [Points: +4]

43. Have you verified that the use of personal information is limited to only what is directly related to delivering a program or service?
Yes [Points: +2]

44. Is the personal information of individuals being used in a decision-making process that directly affects those individuals?
Yes
[Points: +0]

45. Have you verified if the system is using personal information in a way that is consistent with: (a) the current Personal Information Banks (PIBs) and Privacy Impact Assessments (PIAs) of your programs or (b) planned or implemented modifications to the PIBs or PIAs that take new uses and processes into account?
Yes [Points: +0]

46. Please list relevant PIB Bank Numbers.
ICES PIB CBSA PPU 016 and Passage History PIB CBSA PPU 1101

47. What is the highest security classification of the input data used by the system? (Select one)
Protected B [Points: +3]

48. Who controls the data?
Federal government [Points: +1]

49. Will the system use data from multiple different sources?
Yes [Points: +4]

50. Will the system require input data from an Internet- or telephony-connected device? (e.g. Internet of Things, sensor)
No [Points: +0]

51. Will the system interface with other IT systems?
Yes [Points: +4]

52. Who collected the data used for training the system?
Your institution [Points: +1]

53. Who collected the input data used by the system?
Your institution [Points: +1]

54. Please describe the input data collected and used by the system, its source, and method of collection.

The TCI algorithm uses live data from the current passage (such as whether the traveller is travelling alone, the type of ID they have presented, the license plate of the vehicle, etc.) and data from travellers previous trips (i.e. information included in ICES and information from the Passage History database).

The live data is collected when traveller's documents/plate are scanned, and the travel history data is collected from the CBSA Enterprise Data Warehouse (EDW) which contains Passage History data and ICES data. All data derived by the TCI system originates from internal CBSA systems.

About the Data - B. Type of Data

55. Will the system require the analysis of unstructured data to render a recommendation or a decision?
No [Points: 0]

Section 3.2: Mitigation Questions and Answers

Consultations

1. Internal Stakeholders (federal institutions, including the federal public service)
Yes [Points: +1]

2. Which Internal Stakeholders will you be engaging?
Communications services
Strategic Policy and Planning
Data Governance
Program Policy
Legal Services
Other (describe)
Access to Information and Privacy Office
Client Experience / Client Relationship Management
Accessibility Working Groups
TBS Office of the Chief Information Officer

3. Please describe
(Continued)
GBA Plus
Indigenous Affairs Secretariat
Border Services Officers/Superintendents

4. External Stakeholders (groups in other sectors or jurisdictions)
Yes [Points: +1]

5. Which External Stakeholders will you be engaging?
Governments in other Jurisdictions

De-Risking and Mitigation Measures - Data Quality

6. Will you have documented processes in place to test datasets against biases and other unexpected outcomes? This could include experience in applying frameworks, methods, guidelines or other assessment tools.
Yes [Points: +2]

7. Will you be making this information publicly available?
No [Points: +0]

8. Will you be developing a process to document how data quality issues were resolved during the design process?
Yes [Points: +1]

9. Will you be making this information publicly available?
No [Points: +0]

10. Will you undertake a Gender Based Analysis Plus of the data?
Yes [Points: +1]

11. Will you be making this information publicly available?
No [Points: +0]

12. Have you assigned accountability in your institution for the design, development, maintenance, and improvement of the system?
Yes [Points: +2]

13. Will you have a documented process to manage the risk that outdated or unreliable data is used to make an automated decision?
Yes [Points: +2]

14. Will you be making this information publicly available?
No [Points: +0]

15. Will the data used for this system be posted on the Open Government Portal?
No [Points: +0]

De-Risking and Mitigation Measures - Procedural Fairness

16. Will the audit trail identify the authority or delegated authority identified in legislation?
Yes [Points: +1]

17. Will the system provide an audit trail that records all the recommendations or decisions made by the system?
Yes [Points: +2]

18. Will all key decision points be identifiable in the audit trail?
Yes [Points: +2]

19. Will all key decision points within the automated system's logic be linked to the relevant legislation, policy or procedure?
Yes [Points: +1]

20. Will you maintain a log detailing all of the changes made to the model and the system?
Yes [Points: +2]

21. Will the audit trail clearly set out all decision points made by the system?
Yes [Points: +1]

22. Could the audit trail generated by the system be used to help generate a notification of the decision (including a statement of reasons or other notification) where required?
Yes [Points: +1]

23. Will the audit trail identify precisely which version of the system was used for each decision it supports?
Yes [Points: +2]

24. Will the audit trail show who the authorized decision-maker is?
Yes [Points: +1]

25. Will the system be able to produce reasons for its decisions or recommendations when required?
Yes [Points: +2]

26. Will there be a process in place to grant, monitor, and revoke access permission to the system?
Yes [Points: +1]

27. Will there be a mechanism to capture feedback by users of the system?
Yes [Points: +1]

28. Will there be a recourse process planned or established for clients that wish to challenge the decision?
No [Points: +0]

29. Will the system enable human override of system decisions?
Yes [Points: +0]

30. Will there be a process in place to log the instances when overrides were performed?
Yes [Points: +0]

31. Will the audit trail include change control processes to record modifications to the system's operation or performance?
Yes [Points: +2]

32. Will you be preparing a concept case to the Government of Canada Enterprise Architecture Review Board?
Yes [Points: +1]

De-Risking and Mitigation Measures - Privacy

33. If your system uses or creates personal information, will you undertake or have you undertaken a Privacy Impact Assessment, or updated an existing one?
Yes [Points: +1]

34. Please indicate the following in your answer: Title and scope of the Privacy Impact Assessment; How the automation project fits into the program; and Date of Privacy Impact Assessment completion or modification.

The CBSA has committed to submitting the *Traveller Facilitation and Compliance Program Baseline Privacy Impact Assessment* (PIA) to the Office of the Privacy Commissioner and Treasure Board before implementation of the OE project. The PIA for Release 1 of Phase 1of the OE project is forecast for submission August, 2026. A Privacy Protocol Assessment for the Traveller Compliance Indicator Proof of Concept was also completed in 2019.

The automation project fits into the program because it provides improved access to information derived from existing data collected by CBSA to BSOs working in the primary and secondary inspection areas and is expected to provide the likelihood of compliance of travellers. The CBSA will benefit by the improved access to information through improved release/refer decision making, especially in the case of travellers identified as having passage characteristics that are consistent with compliant trends.

35. Will you design and build security and privacy into your systems from the concept stage of the project?
Yes [Points: +1]

36. Will information be used within a closed system (i.e. no connections to the Internet, Intranet or any other system)?
No [Points: +0]

37. If the sharing of personal information is involved, has an agreement or arrangement with appropriate safeguards been established?
No [Points: +0]

38. Will you de-identify any personal information used or created by the system at any point in the lifecycle?
Yes [Points: +0]

All personal information is anonymized before training the TCI models. All personal information is anonymized before scoring a passage. No names, no date of births, no passport IDs and no other personal identifiers are used for scoring/training. These fields are all purged before scoring/training. Only features are used (e.g., number of passages, origin, age group, etc.